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Zero-Knowledge Proofs for Verifiable and Privacy-Preserving SQL Queries in Cloud Databases

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ABSTRACT

Outsourcing relational databases to untrusted cloud providers risks unauthorized data exfiltration and silent query result tampering. We present zk-SQL, a zero-knowledge verifiable query protocol that enables cloud servers to prove the correctness and completeness of SQL results. The system encodes relational algebra operators (SELECT, JOIN, GROUP BY) into arithmetic constraint satisfaction systems (R1CS) using zk-SNARKs. On TPC-H database benchmarks, zk-SQL proves query integrity over 100,000 records in 1.8 seconds with client verification executing in under 15 milliseconds. The primary bottleneck is proof generation overhead for complex nested sub-queries and non-equi joins, which can exhaust server RAM.

Keywords— Cybersecurity, Zero-Knowledge Proofs, Cloud Databases, Privacy-Preserving

INTRODUCTION

Cloud database services require mechanisms ensuring that untrusted hosting providers cannot alter, omit, or inspect database records. Homomorphic encryption handles computation but lacks succinct proofs of query execution correctness.

METHODOLOGY

zk-SQL converts relational tables into cryptographic polynomial commitments (KZG commitments). Query operators produce Groth16 zk-SNARK proofs demonstrating that result tuples satisfy WHERE clauses without revealing unselected rows.

RESULTS

Tested on relational tables up to 100,000 rows, zk-SQL generated succinct proofs (128 bytes) verifiable by lightweight mobile clients in 14.8 ms, verifying complete integrity without server data leaks.

LIMITATIONS

Memory consumption during circuit generation scales quadratically for nested non-equi joins, capping server performance on queries involving multiple Cartesian products.

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